

# Viyom Shukla

LinkedIn: viyom-shukla

GitHub: github.com/viyomshukla

TUF+: 350+ DSA problems | LeetCode: 200+ problems | GFG: 100+ problems

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Location: Noida, India

## EDUCATION

- **Jaypee Institute of Information Technology** Noida, India  
*Integrated B.Tech + M.Tech in Computer Science Engineering; CGPA: 7.62/10* 2022 – 2027
- **Gayatri Public School** Agra, India  
*Class XII (CBSE): 88.6% | Class X (CBSE): 86.6%*

## EXPERIENCE

- **Distance Connect** Remote  
*Web Developer Intern* Aug 2025 – Oct 2025
  - **Platform UI & Architecture**: Spearheaded frontend development for an early-stage EdTech startup connecting students with IIT and AIIMS mentors, comprehensively revamping the user interface to enhance student engagement.
  - **AI Chatbot Integration**: Engineered and deployed an AI-powered conversational interface to automate student onboarding and rapidly resolve queries, significantly reducing manual support overhead.
  - **Real-Time Mentorship Pipeline**: Architected a seamless video consultation flow by integrating the Google Meet API, enabling secure 1:1 virtual mentoring sessions directly within the platform.

## PROJECTS

- **Thumblify — AI-Powered SaaS Platform (Backend: Node.js, MongoDB, REST APIs, Cloudinary)** *Source Code*
  - **AI Generation Engine**: Architected a full-stack SaaS serving 100+ users; generated 3 distinct thumbnail styles in 2K/4K via Gemini and Replicate APIs with fault-tolerant processing.
  - **Secure Authentication**: Implemented Google OAuth 2.0 sign-in with JWT session tokens, safeguarding all user accounts and API endpoints.
  - **Payments & Monetization**: Launched a credit-based subscription model via Razorpay, securing 100% of transactions with signature verification and webhooks.
  - **Conversational Interfaces**: Created 2 conversational channels — an in-app AI chatbot and a WhatsApp bot via Twilio API — enabling single-prompt generation from mobile.
  - **In-Browser Editing**: Embedded TUI Image Editor with 10+ editing tools (crop, filters, text overlays), eliminating reliance on external design software.
  - **Performance & Delivery**: Delivered 1,000+ generated images via Cloudinary CDN and shipped a Chrome extension backed by a low-latency backend.
- **NextTrip — Booking & Management Platform (Backend: Node.js, MongoDB, REST APIs)** *Source Code*
  - **Role-Based Platform**: Engineered a booking platform with Google OAuth 2.0 login and JWT role-based access for 2 roles (admin, traveler), securing 100% of API routes.
  - **Organizer & Identity Module**: Empowered organizers to publish trips with rich image galleries; built an Aadhaar document-upload flow with admin review and manual approval before verifying travelers for KYC-compliant onboarding.
  - **Payments**: Automated booking payments via Razorpay API with server-side verification of each transaction, reducing manual reconciliation to 0.
  - **Admin Control & Data Layer**: Built an admin dashboard for trip listings, promotional imagery, and booking analytics on a normalized 6-collection MongoDB schema with indexed queries.
- **DeepMood — GAN-Based Class-Imbalance Correction (Python, ML)** *Notebook*
  - **Python Data Pipeline**: Built an end-to-end Python pipeline (pandas, scikit-learn) with a GAN that synthesized 5,000+ records to correct severe class imbalance in a skewed dataset.
  - **Model Development & Evaluation**: Benchmarked 4 classifiers on real vs. GAN-augmented data (92% accuracy), evaluating precision, recall, F1-score, and confusion matrix to minimize false negatives.

## RESEARCH & PUBLICATIONS

- **IEEE Publications — Anomaly Detection with Generative Models (2 Papers)**:
  - (1) **Conditional GAN (CGAN) paper** — *Enhancing Network Intrusion Detection Using Conditional GANs for Imbalanced Data Classification*, published on [IEEE Xplore](#): used a Conditional GAN to generate synthetic samples of rare attack types in NSL-KDD, then trained Random Forest, SVM, and Neural Network classifiers on the balanced data to improve detection of underrepresented attacks (99.8% accuracy).
  - (2) **Quantum GAN (QGAN) paper** — *Enhancing Network Intrusion Detection Using Quantum GANs for Imbalanced Data Classification*, accepted at IEEE IC3-2026 (IEEE Xplore, forthcoming): built a Quantum GAN in PyTorch whose generator uses quantum-computing ideas to create realistic samples of rare cyber-attacks, fixing the imbalance in the NSL-KDD dataset; then tested attack detection using three quantum classifiers — Quantum SVM, Quantum k-NN, and a Quantum Decision Tree (built with Qiskit).
- **International Conference Speaker**: Presented published research at the 17th IEEE IC3-2025, IIIT Noida. [Certificate](#)

## SKILLS SUMMARY

- **Languages**: C++ (Proficient), Python, JavaScript, C, SQL
- **Frontend**: React.js, Tailwind CSS, HTML/CSS, Chrome Extensions
- **Backend & APIs**: Node.js, Express.js, REST APIs, Google OAuth 2.0, JWT Authentication, Razorpay, Twilio, Gemini/Replicate APIs
- **Databases, Cloud**: MongoDB Atlas, MySQL, Firebase, AWS (EC2, S3), Cloudinary, Git, GitHub
- **Core CS**: Data Structures & Algorithms, Operating Systems, Computer Networks, DBMS, Distributed & Scalable Systems, OOP, Machine Learning

## ACHIEVEMENTS

- **Competitive Programming**: Solved 450+ DSA and 50+ SQL problems across TUF+, LeetCode, and GeeksforGeeks, focusing on optimization, time complexity, and algorithm design.
- **National Hackathons — SIH'25 Finalist**: Reached the final round of Smart India Hackathon 2025, shortlisted from thousands of national entries; also competed in SIH 2023, Adobe India Hackathon '25 and Amazon ML Challenge.
- **Student Placement Committee Member**: Guided 100+ students through their job search; coordinated placement drives and tracked companies, job roles, and hiring outcomes.